

MSC FINANCE & BIG DATA

Optimization Methods with Applications to Finance • 2627_FI_5_3_073_E

Before you arrive

The five concepts this course assumes

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A word before you begin

This is not optional reading. The course assumes every page of it from the first hour of Session 1, and none of it is taught in class — the ten sessions are fully used by the material they are meant to cover.

Read it, then take the self-test. If the self-test shows that you do not know this material, **it is your responsibility to prepare yourself before we meet.** The references at the end tell you exactly where to go, and an evening or two is normally enough. Students who arrive prepared find this course demanding but entirely doable; students who do not spend the whole course catching up.

What this pack is — and what it deliberately leaves out

This is **not** a summary of the course. It covers only the mathematics the course *assumes* and never stops to teach: **sets, matrices, limits and continuity, derivation, and differential equations**, plus a short probability refresher.

Everything else you will be taught from scratch. **Do not prepare it** — you would only be revising material that Session 1 opens with anyway.

Read this pack	Taught in class — skip
Sets, intervals, $\forall$ and $\exists$	Convex sets and convex functions
Vectors, matrices, transpose, determinant	Quadratic forms, positive definiteness
Limits and continuity	Topology, compactness, Weierstrass
Derivation, one-sided derivatives, C^k	Gradient, Hessian, Taylor expansion
First-order differential equations	Optimality conditions, KKT, Newton
Expectation, variance, covariance	Covariance matrices, portfolio variance
	Normal quantiles and chance constraints
	Linear programming and Excel Solver

Two notational habits to absorb now

1. This course writes the transpose as a *left* superscript: tx , not $x^\top$.
 2. It writes open intervals in the French style: $]0, 1[$, not $(0, 1)$.
- Both appear in the first ten minutes of Session 1. Students who guess at them lose an hour.

How to use it

One or two sittings. Then take the **self-test in Section 7** — eighteen questions, answers in Section 8. Score 14 or above and you are ready; below 10, spend an evening with the references before Session 1.

One practical item: install the Excel **Solver add-in** before the first workshop (Section 6). You will be taught how to use it; you will not be given time to install it.

1 Sets, intervals and logical notation

$]0, 1[$	open interval	endpoints <i>excluded</i> (elsewhere written $(0, 1)$)
$[0, 1]$	closed interval	endpoints included
$\forall$	for all	$\forall x \in S$: “for every x in S ”
$\exists$	there exists	$\nexists$: “there is no”
$\cap, \cup$	intersection, union	
$\subseteq$	subset	
S^c	complement	everything not in S
$\Rightarrow, \iff$	implies, equivalent to	
$\{x : P(x)\}$	set-builder	“the x such that $P(x)$ holds”

Worked example — reading a set definition

$$S = \{(w_1, w_2) \in \mathbb{R}^2 : w_1 + w_2 = 1, w_1 \geq 0, w_2 \geq 0\}.$$

Read it left to right. The pairs (w_1, w_2) of real numbers such that the two numbers sum to one and neither is negative. The colon is “such that”; the commas are “and”.

Test membership.

- $(0.7, 0.3)$: sum = 1 ✓, both ≥ 0 ✓. **In S .**
- $(1.4, -0.4)$: sum = 1 ✓, but $w_2 < 0$. **Not in S .**
- $(0.5, 0.4)$: sum = 0.9 $\neq 1$. **Not in S .**

Picture it. Since $w_2 = 1 - w_1$, the set is the straight segment joining $(1, 0)$ and $(0, 1)$ — a one-dimensional object sitting inside $\mathbb{R}^2$.

Quantify over it.

$$\forall (w_1, w_2) \in S : w_1 \leq 1 \quad \text{and} \quad \exists (w_1, w_2) \in S : w_1 > 0.5.$$

The first says *every* element has the property; the second says *at least one* does. Session 1 will ask you to prove statements of exactly this shape.

2 Matrix calculation

2.1 Vectors and the transpose

An element of $\mathbb{R}^n$ is a *column*; ${}^t x$ turns it into a row:

$$x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}, \quad {}^t x = (x_1, \dots, x_n), \quad {}^t({}^t x) = x.$$

So ${}^t xy = \sum_i x_i y_i$ is a *number*, and ${}^t x Ay$ is a number too.

2.2 The three operations

$$(A + B)_{ij} = a_{ij} + b_{ij}, \quad (AB)_{ij} = \sum_k a_{ik} b_{kj}, \quad \det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc.$$

For AB to exist, the columns of A must match the rows of B . And $AB \neq BA$ in general.

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \quad \text{when } \det A \neq 0.$$

Worked example — the three shapes you must not confuse

Let

$$x = \begin{pmatrix} 0.5 \\ 0.3 \\ 0.2 \end{pmatrix}, \quad y = \begin{pmatrix} 4 \\ 9 \\ 12 \end{pmatrix}, \quad A = \begin{pmatrix} 4 & 2 & 1 \\ 2 & 25 & 12 \\ 1 & 12 & 64 \end{pmatrix}.$$

(a) ${}^t xy$ — a **number**.

$${}^t xy = 0.5(4) + 0.3(9) + 0.2(12) = 2.0 + 2.7 + 2.4 = \mathbf{7.1}$$

(b) Ax — a **vector**. One row at a time:

$$\begin{array}{ll} \text{row 1 : } 4(0.5) + 2(0.3) + 1(0.2) = 2.0 + 0.6 + 0.2 = 2.8 \\ \text{row 2 : } 2(0.5) + 25(0.3) + 12(0.2) = 1.0 + 7.5 + 2.4 = 10.9 \\ \text{row 3 : } 1(0.5) + 12(0.3) + 64(0.2) = 0.5 + 3.6 + 12.8 = 16.9 \end{array} \quad Ax = \begin{pmatrix} 2.8 \\ 10.9 \\ 16.9 \end{pmatrix}$$

(c) ${}^t x Ax$ — a **number again**. Take the inner product of x with the vector from (b):

$${}^t x(Ax) = 0.5(2.8) + 0.3(10.9) + 0.2(16.9) = 1.40 + 3.27 + 3.38 = \mathbf{8.05}$$

The habit worth forming. Before computing anything, ask what *shape* the answer should be. Number, vector or matrix? Expressions of the form ${}^t x Ax$ recur throughout the course, and confusing (b) with (c) is the most common error in the first workshop.

And order matters. With $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, $B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$:

$$AB = \begin{pmatrix} 2 & 1 \\ 4 & 3 \end{pmatrix} \quad \text{but} \quad BA = \begin{pmatrix} 3 & 4 \\ 1 & 2 \end{pmatrix}.$$

3 Limits and continuity

f is **continuous** at x_0 if $f(x)$ approaches $f(x_0)$ as x approaches x_0 :

$$\forall \varepsilon > 0, \exists \delta > 0 : |x - x_0| < \delta \implies |f(x) - f(x_0)| < \varepsilon,$$

equivalently: $x_n \rightarrow x_0$ implies $f(x_n) \rightarrow f(x_0)$ for every sequence.

Worked example — a continuity proof, in full

Show that $f(x) = x^2$ is continuous at $x_0 = 2$.

Step 1 — factorise the quantity to be controlled.

$$|x^2 - 4| = |x - 2| |x + 2|.$$

The first factor is what we control; the second is a nuisance to be bounded.

Step 2 — bound the nuisance. Agree in advance to stay within 1 of 2. Then $|x - 2| < 1$ gives $1 < x < 3$, hence $|x + 2| < 5$, and so

$$|x - 2| < 1 \implies |x^2 - 4| < 5|x - 2|.$$

Step 3 — choose δ . Given $\varepsilon > 0$, take

$$\delta = \min\left(1, \frac{\varepsilon}{5}\right).$$

Then $|x - 2| < \delta$ gives $|x^2 - 4| < 5\delta \leq \varepsilon$. □

Check it with numbers. For $\varepsilon = 0.1$ the recipe gives $\delta = 0.02$. At $x = 2.02$: $|x^2 - 4| = 0.0804 < 0.1$ ✓.

Read the structure. ε is the tolerance *demand*ed; δ is the tolerance you *supply*. δ is allowed to depend on ε — that is the whole content of the definition.

4 Derivation

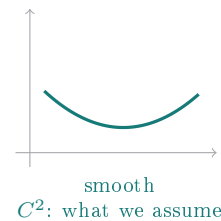
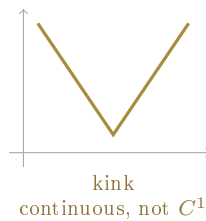
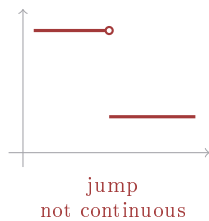
4.1 The derivative

$$f'(x_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h},$$

when the limit exists — that is, when the left and right limits agree.

Remember

Differentiable $\Rightarrow$ continuous. The converse is false. A *kink* is precisely a point of continuity without differentiability.



Worked example — a kink you already know

The payoff at maturity of a call option struck at $K = 100$ is

$$f(S) = \max(S - 100, 0) = \begin{cases} 0 & S \leq 100 \\ S - 100 & S > 100. \end{cases}$$

Continuous at $S = 100$? From the left the value tends to 0; from the right, $S - 100 \rightarrow 0$; and $f(100) = 0$. All three agree. **Yes.**

Differentiable at $S = 100$? Compute both one-sided difference quotients:

$$\lim_{h \rightarrow 0^-} \frac{f(100 + h) - f(100)}{h} = \frac{0 - 0}{h} = 0, \quad \lim_{h \rightarrow 0^+} \frac{f(100 + h) - f(100)}{h} = \frac{h - 0}{h} = 1.$$

Slopes 0 and 1: they disagree, so **not differentiable at the strike**.

The same happens for $|x|$ at 0, and for $\max\{f, g\}$ wherever f and g cross.

4.2 Smoothness classes

$$C^0 \supset C^1 \supset C^2$$

C^0 : continuous. C^1 : f' exists and is continuous. C^2 : f'' exists and is continuous.

Where you will need this

The course assumes $f \in C^2$ from the second day onwards. That single phrase is what licenses everything built later. When you meet it in a lecture, you should hear: “*we may safely differentiate twice*”. Where it fails — option payoffs, $|x|$, the ℓ_1 norm — the standard methods do not apply, and the course says so explicitly.

4.3 Lipschitz continuity

f is **Lipschitz** with constant L if

$$|f(x) - f(y)| \leq L|x - y| \quad \text{for all } x, y,$$

a bound on how fast f can change. Lipschitz $\Rightarrow$ continuous. You will meet the phrase in one theorem's hypothesis; you need only recognise it.

5 Differential equations

5.1 Definitions

An **ordinary differential equation** relates an unknown function to its derivatives; its *order* is the highest derivative present. With a starting value it becomes an **initial value problem**:

$$\frac{dy}{dt} = F(t, y), \quad y(t_0) = y_0.$$

5.2 The two you need

Separable. If $\frac{dy}{dt} = g(t)h(y)$, separate and integrate: $\int \frac{dy}{h(y)} = \int g(t) dt$.

Linear, constant coefficients. For $y' + ay = b$ with $a \neq 0$:

$$y(t) = \frac{b}{a} + \left(y_0 - \frac{b}{a}\right)e^{-at},$$

which moves from y_0 to the steady state b/a at exponential speed.

Worked example 1 — the equation everything reduces to

$$\frac{dy}{dt} = ay, \quad y(0) = y_0.$$

Separate: $\int \frac{dy}{y} = \int a dt$, so $\ln y = at + c$ and

$$y(t) = y_0 e^{at}$$

Growth if $a > 0$, decay if $a < 0$.

Application — €1,000 at 5%, credited continuously. Interest proportional to the balance means $\frac{dV}{dt} = 0.05V$, so $V(t) = 1000e^{0.05t}$ and

$$V(3) = 1000e^{0.15} = \mathbf{1,161.83}.$$

Compare the same nominal rate compounded discretely:

annual, $1000(1.05)^3$	€1,157.63
monthly, $1000\left(1 + \frac{0.05}{12}\right)^{36}$	€1,161.47
continuous, $1000e^{0.15}$	€1,161.83

They converge because $\left(1 + \frac{r}{m}\right)^m \rightarrow e^r$. This is also why *log* returns add across periods while simple returns do not.

Worked example 2 — solve, then always verify

Solve $y' + 2y = 6$ with $y(0) = 1$.

Here $a = 2$, $b = 6$, steady state $b/a = 3$, so

$$y(t) = 3 + (1 - 3)e^{-2t} = \mathbf{3 - 2e^{-2t}}.$$

Verify by substitution. $y' = 4e^{-2t}$, hence

$$y' + 2y = 4e^{-2t} + 2(3 - 2e^{-2t}) = 4e^{-2t} + 6 - 4e^{-2t} = 6 \checkmark, \quad y(0) = 3 - 2 = 1 \checkmark.$$

Read the behaviour. y rises from 1 towards 3; at $t = 1$ it is $3 - 2e^{-2} = 2.73$. The exponential term dies away, so the system forgets where it started. **Every linear first-order system behaves this way.**

Where you will need this

The algorithms taught in this course are *discretisations of differential equations*. The connection is made in class; you need only be comfortable with $\dot{y} = ay$ and its solution when it appears.

6 Probability refresher, and one practical item

6.1 Expectation, variance, covariance

$$\begin{aligned}\mathbb{E}[X] &= \sum_i p_i x_i, & \text{Var}(X) &= \mathbb{E}[X^2] - (\mathbb{E}X)^2, & \sigma_X &= \sqrt{\text{Var}(X)}, \\ \mathbb{E}[aX + b] &= a\mathbb{E}[X] + b, & \text{Var}(aX + b) &= a^2 \text{Var}(X), \\ \text{Cov}(X, Y) &= \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y], & \rho_{XY} &= \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} \in [-1, 1].\end{aligned}$$

Worked example — from a scenario table to a covariance

	Boom	Normal	Recession
probability p_s	0.3	0.5	0.2
X	12	6	-4
Y	6	12	-8

Means. $\mathbb{E}[X] = 0.3(12) + 0.5(6) + 0.2(-4) = 3.6 + 3.0 - 0.8 = \mathbf{5.8}$; $\mathbb{E}[Y] = 1.8 + 6.0 - 1.6 = \mathbf{6.2}$.

Variances. $\mathbb{E}[X^2] = 0.3(144) + 0.5(36) + 0.2(16) = 64.4$, so $\text{Var}(X) = 64.4 - 5.8^2 = \mathbf{30.76}$ and $\sigma_X = 5.55$.

$\mathbb{E}[Y^2] = 0.3(36) + 0.5(144) + 0.2(64) = 95.6$, so $\text{Var}(Y) = 95.6 - 6.2^2 = \mathbf{57.16}$ and $\sigma_Y = 7.56$.

Covariance and correlation. $\mathbb{E}[XY] = 0.3(72) + 0.5(72) + 0.2(32) = 64.0$, so

$$\text{Cov}(X, Y) = 64.0 - (5.8)(6.2) = \mathbf{28.04}, \quad \rho = \frac{28.04}{(5.5462)(7.5604)} = \mathbf{0.669}.$$

Common slip

$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$ **only when X and Y are uncorrelated**. In general

$$\text{Var}(aX + bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2ab \text{Cov}(X, Y).$$

The cross term is the single most consequential quantity in the entire course. Drop it and half of what you are about to learn stops working.

6.2 Install Excel Solver before day one

Solver ships with Excel but is not enabled by default. You will be taught how to use it; you will not be given class time to install it.

Windows: File $\rightarrow$ Options $\rightarrow$ Add-ins $\rightarrow$ *Manage: Excel Add-ins* $\rightarrow$ Go $\rightarrow$ tick **Solver Add-in** $\rightarrow$ OK. It then appears under the **Data** tab, far right.

macOS: Tools $\rightarrow$ Excel Add-ins $\rightarrow$ tick **Solver**.

Confirm you can see the Solver button before Session 2. That is all that is required of you.

7 Self-test

Eighteen questions. Answers overleaf.

1. Let $S = \{(w_1, w_2) : w_1 + w_2 = 1, w_1 \geq 0, w_2 \geq 0\}$. Is $(0.7, 0.3) \in S$? Is $(1.4, -0.4) \in S$? Justify each.
2. What is the difference between $]0, 1[$ and $[0, 1]$?
3. Write in symbols: “for every x in S there exists $\varepsilon > 0$ such that $x + \varepsilon \in S$ ”.
4. Compute ${}^t x y$ for $x = (1, 2, 3)$ and $y = (2, 0, -1)$.
5. With $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, compute AB and BA .
6. Compute $\det \begin{pmatrix} a & b \\ b & c \end{pmatrix}$, and give A^{-1} for $A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$.
7. For $A = \begin{pmatrix} 4 & 2 \\ 2 & 5 \end{pmatrix}$ and $x = (0.6, 0.4)$, compute Ax and then ${}^t x Ax$. State the shape of each.
8. Give the ε - δ definition of continuity at x_0 .
9. Prove $f(x) = x^2$ is continuous at $x_0 = 2$ by exhibiting a δ for a given ε .
10. Is $f(x) = |x|$ continuous at 0? Differentiable at 0? Justify both.
11. For $f(S) = \max(S - 100, 0)$, compute the left and right derivatives at $S = 100$.
12. What does “ f is C^2 ” mean?
13. State the definition of a Lipschitz function.
14. Solve $\frac{dy}{dt} = 3y$ with $y(0) = 2$.
15. An account of €1,000 earns 5% compounded continuously. Write the differential equation and find $V(3)$.
16. Solve $y' + 2y = 6$ with $y(0) = 1$, and verify your answer by substitution.
17. X takes 12, 6, -4 with probabilities 0.3, 0.5, 0.2. Find $\mathbb{E}[X]$ and $\text{Var}(X)$.
18. State $\text{Var}(aX + bY)$ in full. Under what condition does the cross term vanish?

8 Answers

1. $(0.7, 0.3)$: sum = 1 and both ≥ 0 , so **yes**. $(1.4, -0.4)$: sum = 1 but $w_2 < 0$, so **no**.
2. $]0, 1[$ excludes 0 and 1; $[0, 1]$ includes them. Session 1 turns this into the distinction between *open* and *closed* sets.
3. $\forall x \in S, \exists \varepsilon > 0 : x + \varepsilon \in S$.
4. $1(2) + 2(0) + 3(-1) = -1$.
5. $AB = \begin{pmatrix} 2 & 1 \\ 4 & 3 \end{pmatrix}$, $BA = \begin{pmatrix} 3 & 4 \\ 1 & 2 \end{pmatrix}$. They differ: matrix multiplication does not commute.
6. $ac - b^2$. $\det A = 1$, so $A^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}$.
7. $Ax = (4(0.6) + 2(0.4), 2(0.6) + 25(0.4)) = (3.2, 11.2)$ — a *vector*. ${}^t x Ax = 0.6(3.2) + 0.4(11.2) = 1.92 + 4.48 = 6.40$ — a *number*.
8. $\forall \varepsilon > 0, \exists \delta > 0 : |x - x_0| < \delta \Rightarrow |f(x) - f(x_0)| < \varepsilon$.
9. $|x^2 - 4| = |x - 2||x + 2|$; if $|x - 2| < 1$ then $|x + 2| < 5$, so take $\delta = \min(1, \varepsilon/5)$.
10. Continuous: yes. Differentiable at 0: **no** — the left derivative is -1 , the right $+1$.
11. Left derivative 0, right derivative 1. Not differentiable at the strike.
12. f, f' and f'' all exist and are continuous.
13. $|f(x) - f(y)| \leq L|x - y|$ for all x, y , for some constant L .
14. Separable: $\int dy/y = \int 3 dt$, giving $y(t) = 2e^{3t}$.
15. $\frac{dV}{dt} = 0.05V$, $V(0) = 1000$, so $V(t) = 1000e^{0.05t}$ and $V(3) = 1000e^{0.15} = 1,161.83$.
16. $y(t) = 3 - 2e^{-2t}$. Check: $y' = 4e^{-2t}$ and $y' + 2y = 6$ ✓, $y(0) = 1$ ✓.
17. $E[X] = 5.8$; $E[X^2] = 64.4$; $\text{Var}(X) = 64.4 - 33.64 = 30.76$.
18. $\text{Var}(aX + bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2ab \text{Cov}(X, Y)$. The cross term vanishes when $\text{Cov}(X, Y) = 0$, i.e. when X and Y are uncorrelated.

Scoring

14–18 — ready. **10–13** — revise the sections you missed. **below 10** — work through the references before Session 1.

Questions 7, 12 and 18 matter most: the shape of ${}^t x Ax$, what C^2 means, and the covariance cross term. Those three are assumed on every day of the course.

9 References

To revise

- Simon, C. & Blume, L., *Mathematics for Economists*, Norton. Chapters 6–8 for calculus and continuity, 8–9 for linear algebra, 24–25 for differential equations. The closest match to this course.

The course texts (for reference during the course — not before it)

- Palomar, D. P., *Portfolio Optimization*, Chapman & Hall/CRC, 2024.
- Cornuéjols, G., Peña, J. & Tütüncü, R., *Optimization Methods in Finance*, 2nd ed., Cambridge, 2018.

See you in Session 1.